

resulting in lower TIPS yields and a higher perceived break-even inflation rate. Rather than waiting to buy TIPS until everyone starts worrying about inflation, a natural time to purchase is before inflation becomes a major concern.

Despite the other factors of TIPS pricing, the difference between Treasury and TIPS rates for the same maturity represents a reasonable market estimate of future inflation expectations. Exhibit 7.6 returns us to January 3, 2017, to find this difference.

Consider the thirty-year maturity. A thirty-year Treasury bond provides its owner a nominal 3.04 percent yield. Its real yield is unknown and depends on realized future inflation. Meanwhile, a thirty-year TIPS offers a real yield of 0.94 percent. Its nominal yield is unknown, as it also depends on realized future inflation. The difference between these yields is the implied break-even inflation rate: 2.1 percent. Without a liquidity or inflation-protection premium, this represents the market's equilibrium estimate of future inflation. Over the next thirty years, the markets have priced in expectations for inflation of only around 2 percent.

If realized annual inflation exceeds 2.1 percent over the next thirty years, then TIPS will outperform Treasuries. But if it falls short, TIPS will underperform. If enough traders thought inflation would be higher than this, they would *buy* TIPS and *sell* Treasuries, raising the price of TIPS, and giving us lower TIPS yields, higher Treasury yields, and a larger break-even inflation rate. Such trading would continue until the market reaches the equilibrium we observe.

Exhibit 7.6

US Government Yield Curve for Constant Maturity Bonds, January 3, 2017

Maturity	Treasury Yield	TIPS Yield	Implied Break- Even Inflation Rate
1 Year	0.89%	-0.91%	1.80%